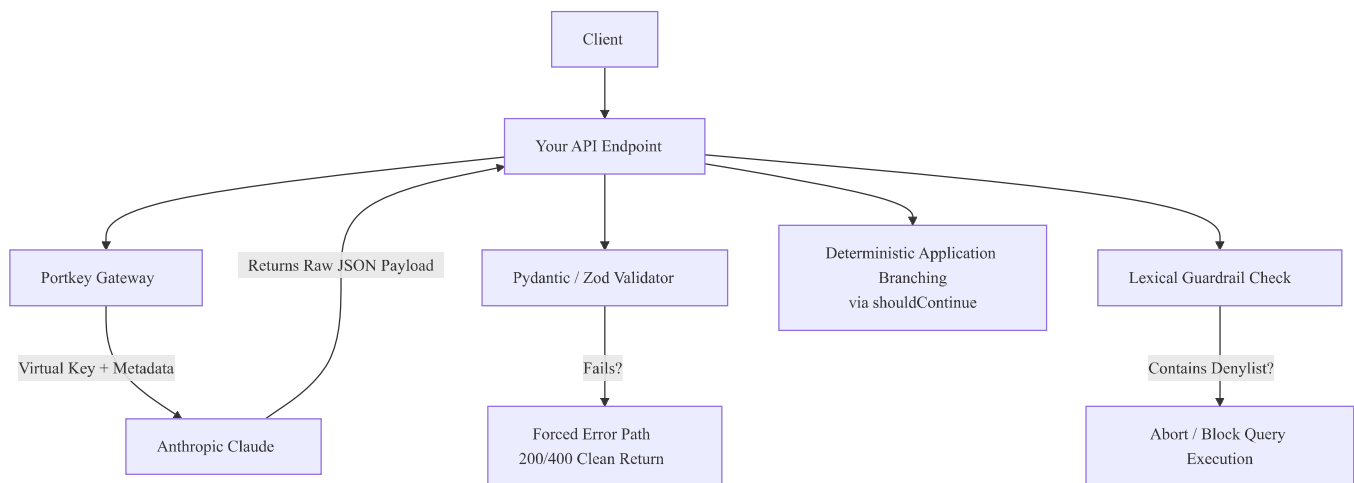


Week 1 - Backend AI Engineering Assignment

Main Specs and Requirements



Core idea: CTF Research Copilot API

The idea: a user sends a question like:

“Find my notes about APK dynamic analysis and summarize the recommended workflow.”

Your API calls **Claude through Portkey**, lets Claude use **2–3 tools over your local data**, validates every tool input/output, forces a **schema-valid final response**, and blocks any dangerous model-generated SQL/query before execution.

This fits the assignment perfectly because it demonstrates the four required patterns: gateway abstraction, structured output, tool factory with context, and generated-query guardrails.

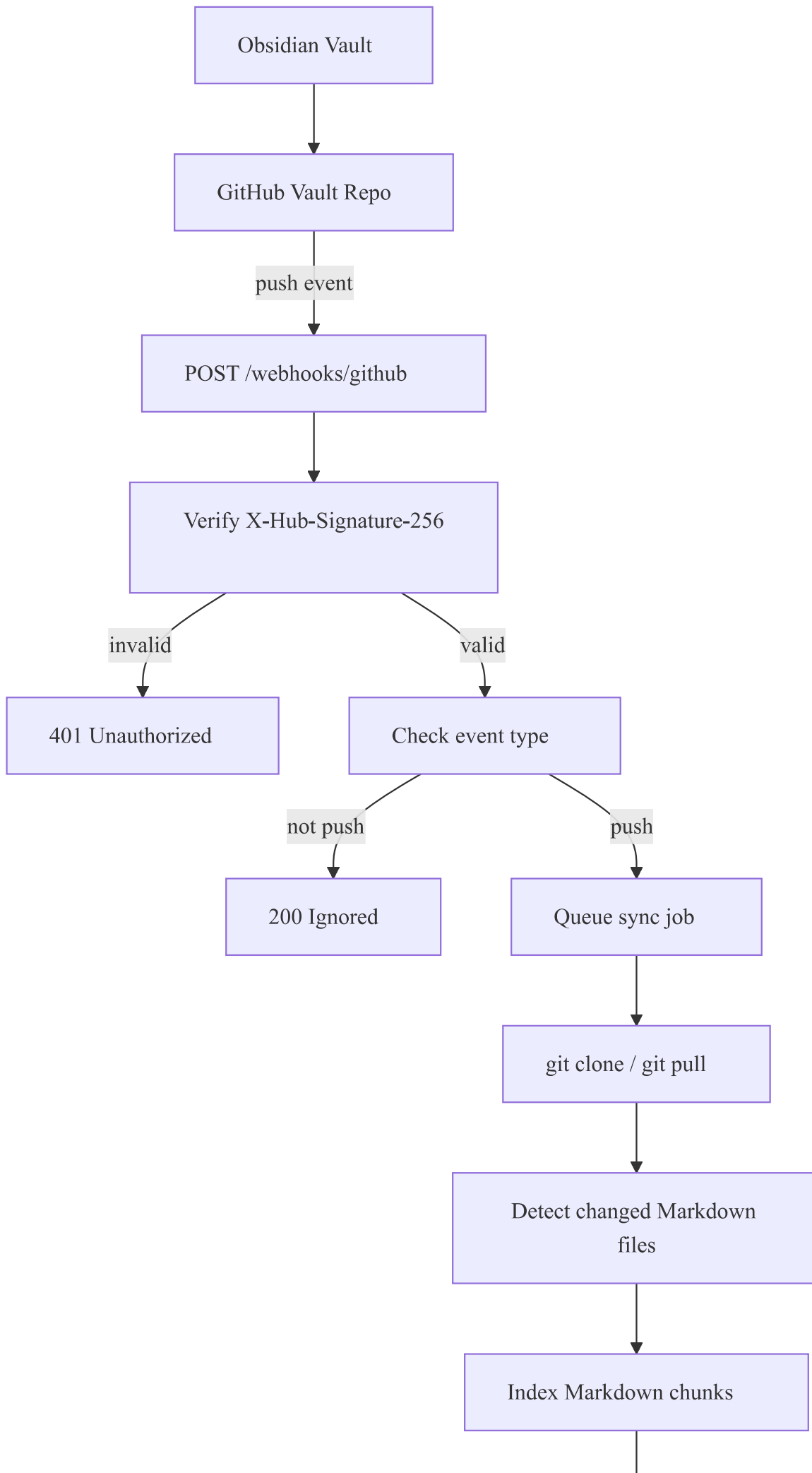
This work is also inspired from the recent rumoured leaked info from Anthropic, revealing the use of thousands of linked notes that organize research, documentation and internal AI concepts, with this project we can leverage this idea, but on a smaller scale, as the notes taking during previous CTF competitions and trainings are very useful in solving new challenges, mimicking a sort of human behaviour, or more accurately, using a "second brain" to optimize the solve rate and timing.

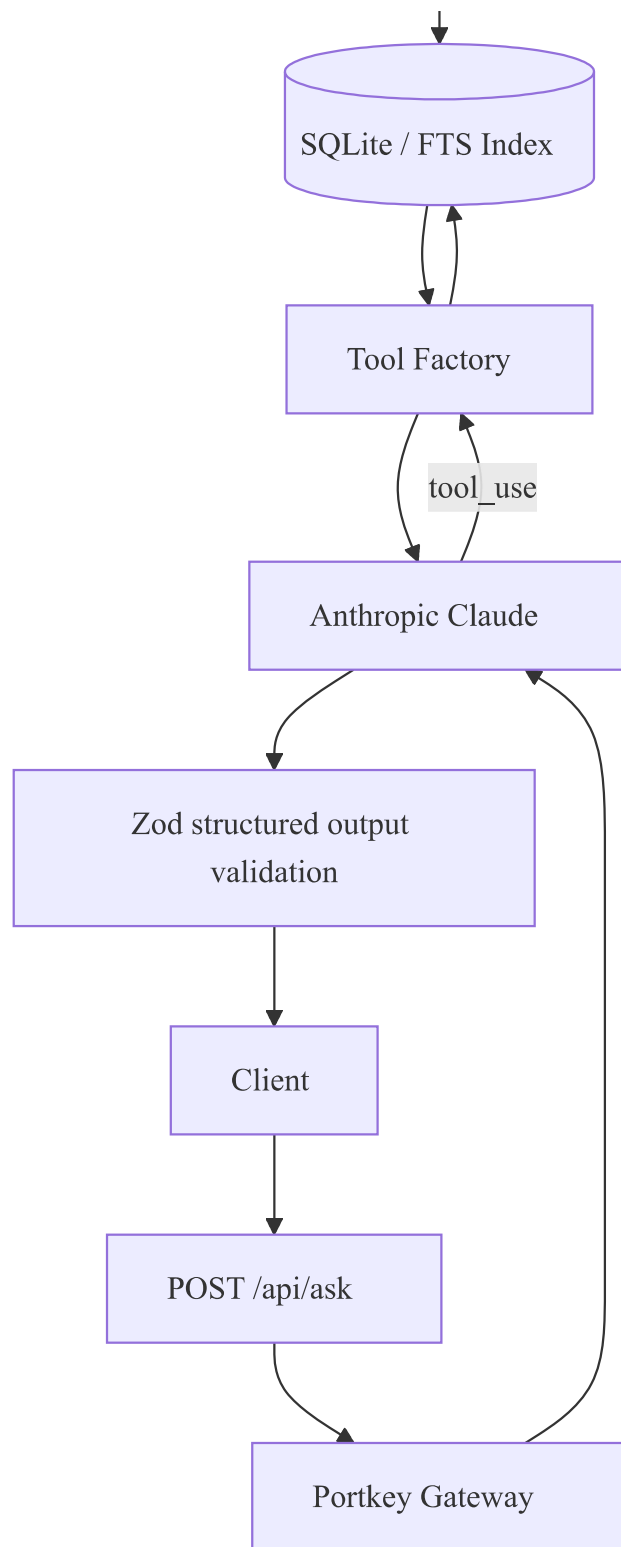
Why we chose syncing Obsidian notes from Github ?

We all know that Obsidian is the leading note taking app available on the market, and the hosting of vaults online for sync is fairly doable, but it still requires a paid subscription.

The CEO of Obsidian itself proposed some free ways to sync in a Reddit comment, and that inspired me to try using that for this project, as well as my extensive use of it and the decent amount of notes and writeups I published so far.

The idea also comes from the fact that solves are most of the time a collaboration of all the team members, and considering the delegation of tasks by categories, makes syncing everything into an organization GitHub repo more optimized and suitable for collaborative team work.





BUILD ORDER

1. Fastify backend
2. GitHub webhook endpoint
3. Signature verification
4. Sync job queue
5. Vault clone / pull
6. Markdown changed-file detection
7. SQLite indexing
8. `/api/index/status`
9. `/api/ask` with Claude + Portkey
10. Tools over indexed notes

The service is provider-swappable through Portkey. The submitted demo uses Gemini because Claude credits were unavailable, but the code path supports swapping to Claude by changing AI_MODEL to @anthropic-slug/claude-...

Testing checklist

You need these tests:

Webhook security:

- ✓ accepts valid GitHub signature ✓ 2026-07-08
- ✓ rejects invalid signature ✓ 2026-07-08
- ✓ rejects missing signature ✓ 2026-07-08
- ✓ ignores non-push event ✓ 2026-07-08

Sync:

- ✓ clones repo if vault cache does not exist ✓ 2026-07-08
- ✓ pulls repo if vault cache exists ✓ 2026-07-08
- ✓ indexes only .md files ✓ 2026-07-08
- ✓ ignores .obsidian and attachments ✓ 2026-07-08

Indexing:

- ✓ chunks Markdown by headings ✓ 2026-07-08
- ✓ skips unchanged files by hash ✓ 2026-07-08

☒ updates changed files ☒ 2026-07-08

AI gate later:

☐ final Claude output validates with Zod

☐ forced error path works

☒ malicious generated SQL is blocked (SQLi) ☒ 2026-07-08

- Fun fact: As my last name abbreviation is SQL and i'm the Web Exploitation player in my ctf team, my nickname is SQL injection

Best MVP target

For your next milestone, we will try to focus on building only this:

```
* POST /webhooks/github | GET /api/index/status | POST /api/syncSQLite  
* Markdown index
```

Then prove:

```
I push to my Obsidian vault GitHub repo. GitHub calls my webhook. My service  
verifies the signature. My service pulls the latest vault. My service indexes  
Markdown notes. The status endpoint shows updated documents/chunks.
```

After that, adding Claude + Portkey tools becomes clean because the data layer is already alive.

The screenshot shows a 'Recent Deliveries' tab in a dark-themed interface. A delivery is listed with a green checkmark, a hexagonal icon, the ID 'a6c5db3c-7ae5-11f1-8c22-cbb3418c8511', and the event 'ping'. The timestamp is '2026-07-08 16:57:06'. Below this, the 'Request' tab is selected, showing a 'Response' of '200' in a green box. A 'Redeliver' button and a completion time of '6.32 seconds' are visible. The 'Headers' section is expanded, displaying the following details:

- Request URL: https://5873-196-119-123-139.ngrok-free.app/webhooks/github
- Request method: POST
- Accept: */*
- Content-Type: application/json
- User-Agent: GitHub-Hookshot/29e815f
- X-Github-Delivery: a6c5db3c-7ae5-11f1-8c22-cbb3418c8511
- X-Github-Event: ping
- X-Github-Hook-Id: 650641164

webhook ping is good

testing the API is good

```
> ctf-copilot-api@1.0.0 test
> vitest run

RUN v2.1.9 /mnt/c/Users/saied/ctf-copilot-api

✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
> tests/globalWebhookRoute.test.ts (4)
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
> tests/globalWebhookRoute.test.ts (4)
  > GitHub webhook route (4)
    ✓ accepts GitHub ping deliveries
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
> tests/globalWebhookRoute.test.ts (4)
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
> tests/globalWebhookRoute.test.ts (4)
  > GitHub webhook route (4)
    ✓ accepts GitHub ping deliveries
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
> tests/globalWebhookRoute.test.ts (4)
  > GitHub webhook route (4)
    ✓ accepts GitHub ping deliveries
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
✓ tests/globalWebhookRoute.test.ts (4) 27134ms
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
✓ tests/globalWebhookRoute.test.ts (4) 27134ms
✓ tests/globalSchemasAndGuardrails.test.ts (22)
> tests/globalAskAndHealthRoutes.test.ts (3)
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
✓ tests/globalWebhookRoute.test.ts (4) 27134ms
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
✓ tests/globalWebhookRoute.test.ts (4) 27134ms
✓ tests/globalSchemasAndGuardrails.test.ts (22)
> tests/globalAskAndHealthRoutes.test.ts (3)
✓ tests/globalAskAndHealthRoutes.test.ts (3) 15716ms
✓ tests/globalConfig.test.ts (4)
✓ tests/globalGithubSignature.test.ts (4)
✓ tests/globalIngestionAndTools.test.ts (10) 1195ms
✓ tests/globalSchemasAndGuardrails.test.ts (22)
✓ tests/globalSyncQueue.test.ts (1)
✓ tests/globalWebhookRoute.test.ts (4) 27134ms

Test Files 7 passed (7)
Tests 48 passed (48)
Start at 17:25:22
Duration 132.90s (transform 3.48s, setup 0ms, collect 38.78s, tests 44.13s, environment 2ms, prepare 36.25s)
```

Sync: ok

```
aya@Laptop: /mnt/c/Users/saied/ctf-copilot-api$ curl -X POST http://localhost:3000/api/sync
{"ok":true,"source":"manual","commitSha":"c5d264d91124c1f71dc534b8e8929a01aa715e9e","changedFiles":["Active Directory/ACL ( Access Co
ntrol List).md","CVEs/CVE-2020-7384.md","CVEs/CVE-2022-24439 PoC.md","CVEs/CVE-2024-28397.md","CVEs/CVE-2025-24071.md","Challenges So
lutions/CDNio solve.md","Challenges Solutions/Compressor HTB.md","Challenges Solutions/Dark Runes - Web Challenge - HTB.md","Challeng
es Solutions/Locked Away HTB.md","Challenges Solutions/Lucky Dice HTB.md","Challenges Solutions/Micro Storage HTB.md","Challenges Sol
utions/Not Positive solve.md","Challenges Solutions/Shiny Hunter HTB.md","Challenges Solutions/xorxorxorxor HTB.md","Cheatsheets/Met
erpreter.md","Cheatsheets/More Reverse Shells.md","Cheatsheets/Reverse Shells.md","Cheatsheets/SQL Injection.md","Cheatsheets/SQLMap.
md","Cheatsheets/XSLT Injection/Attack Techniques.md","Cheatsheets/XSLT Injection/Common Attack Vectors.md","Cheatsheets/XSLT Injecti
on/Evasion Techniques.md","Cheatsheets/XSLT Injection/Real-world Exploitation Examples.md","Cheatsheets/XSLT Injection/XSLT Processor
-Specific Attacks.md","Cheatsheets/cURL.md","Cheatsheets/msfvenom Payloads.md","Machines Solutions/Conversor - HTB.md","Machines Solu
tions/Data (HTB Machine)/CVE-2021-43798.md","Machines Solutions/Eighteen HTB.md","Machines Solutions/Enigma - HTB.md","Machines Solu
tions/Facts HTB.md","Machines Solutions/HackNet Box - HTB.md","Machines Solutions/Imagery HTB.md","Machines Solutions/Pterodactyl HTB.
md","Machines Solutions/Signed HTB (Windows Machine).md","Machines Solutions/Silentium - Privilege Escalation.md","Machines Solutions
/TensorFlow RCE - Artificial HTB.md","Miscellaneous Techniques/NFS.md","Miscellaneous Techniques/SETUID Script.md","Miscellaneous Tec
hniques/Tmux Session Abuse.md","OverTheWire/Bandit Writeup.md","OverTheWire/Leviathan Writeup.md","OverTheWire/Natas Writeup.md","Pri
vilege Escalation/Automated Enumeration Tools.md","Privilege Escalation/Escaping Restricted Shells.md","Privilege Escalation/Kernel E
xploit - Dirty COW PoC.md","Privilege Escalation/LD_PRELOAD Exploitation.md","Privilege Escalation/Linux Capabilities.md","Privilege
Escalation/Linux Privilege Escalation Commands - HTB Academy Cheasheet.md","Privilege Escalation/Screen Privilege Escalation PoC.md",
"Privilege Escalation/Windows Privilege Escalation.md","Privilege Escalation/logrotten Privilege Escalation.md","README.md","Random/3
DES Decryption.md","Random/Brute Forcing.md","Random/BurpSuite Intruder Simulation.md","Random/Cache Poisoning.md","Random/Cookie Ste
aling Via Stored XSS.md","Random/Kubernetes.md","Random/PHP Unserialization for RCE - POP Restaurant HTB Challenge.md","Random/Revers
e Shell.md","Random/Topology HTB (retired).md","Random/XML External Entity Attack Payload.md","Random/grafana2 hash.md","Screens.md"]
aya@Laptop: /mnt/c/Users/saied/ctf-copilot-api$ ":0,"http://localhost:3000/api/index/status
{"documents":66,"chunks":278,"lastIndexedAt":"2026-07-08T16:45:49.149Z","latestCommitSha":"c5d264d91124c1f71dc534b8e8929a01aa715e9e"}
```